Pilot Study For Seagrass Habitat Restoration in West Bay, St. Andrew Bay, Florida

Revised Final Report, February 2022

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Background

Gulf coast estuaries are known for high biological diversity and clear, saline waters that support luxuriant seagrass growth. St. Andrew Bay contains approximately 4,850 hectares of seagrass, the second highest seagrass coverage among the six estuaries in the region (Yarbro & Carlson 2016). Cuban shoal grass (*Halodule wrightii*) and turtle grass (*Thalassia testudinum*) are the most common seagrass species followed by widgeon grass (*Ruppia maritima*), manatee grass (*Syringodium filiforme*), and star grass (*Halophila engelmanni*), in lower densities (Fitzhugh et al. 2016; Yarbro & Carlson 2016).

While St. Andrew Bay as a whole supports healthy seagrass habitat and high species diversity (~3,600 species; Keppner & Keppner 2001), the western arm of St. Andrew Bay known as West Bay has had a long history of anthropogenic impacts to seagrasses. These include construction of the Gulf Intracoastal Waterway (GIWW) in 1938, commercial shrimp aquaculture throughout the 1970s, and repeated freshwater effluent discharges from 1971 to 2011. The corresponding effects on water quality likely resulted in the loss of more than 700 hectares of seagrass at West Bay during the last half century (Brim & Handley, 2007). However, the eventual elimination of shrimp aquaculture operations and wastewater discharge improved water quality and associated light availability (surface irradiance >24%; https://myfwc.com/research/habitat/seagrasses/projects/roadblocks/), once again creating conditions suitable for seagrass growth. Recent work in the area (Figure 1), has demonstrated sufficient light penetration (Appendix 1), and seagrasses, primarily Cuban shoal grass, have begun to recruit within the historically vegetated seagrass habitat areas of West Bay (http://geodata.myfwc.com/datasets/seagrass-habitat-in-florida).

The Florida Fish and Wildlife Conservation Commission (FWC) has designated a seagrass restoration site just south of recent increases in seagrass coverage (Figure 1). Over the course of 2015-2019, the FWC installed approximately two hectares of oyster reef habitat along the deep edge of historic seagrass coverage. At its maximum extent (circa 1964), seagrasses extended from just waterward of mean low water (MLW) to approximately mean low, low water (MLLW; 4-feet NAVD88). Oyster reef habitat was installed to mitigate wind-driven wave energy, reduce suspended sediments, increase water filtration, and further facilitate both the natural recruitment of seagrasses and the expansion of transplanted seagrass beds. To better understand the restoration potential for seagrasses in West Bay, the FWC implemented a series of research studies looking at light availability, donor bed suitability, and transplanting methodology.

While previous research by Fitzhugh et al. (2016) found improved light conditions in St. Andrew Bay (K_{par} = 0.504), light availability was still lowest in West Bay (K_{par} = 0.619). To better understand the light field in the restoration project area, the FWC monitored photosynthetically active radiation (PAR) at six fixed stations, all within historically vegetated areas along the western shore of West Bay. Monitoring was completed along the 1-meter depth contour at 0.1 m, 0.25 m, 0.5 m, and 0.75 meters below the surface and just above the substrate biweekly from May through October 2017. The PAR data gathered from this effort were used to calculate the light extinction coefficient (Kd) and percent surface irradiance. Results suggested that light availability was sufficient for seagrass recruitment and growth to a depth of approximately 1 meter (Appendix 1). Based on this study, the FWC developed a work plan to transplant Cuban shoal grass from nearby donor beds to areas of the project site that had the highest measured light penetration.

In 2018, nine donor sites in the West Bay system were surveyed as potential sources of donor material (Appendix 2, Figure 1). Multiple transects were established at each site and in-water surveys were performed to estimate the abundance of Cuban shoal grass, turtle grass, widgeon grass, and macroalgae. Starting from the seaward edge of each seagrass bed, 1-m^2 quadrats were used to collect measurements along a 12.20-meter-wide (40 ft.) belt transect every 15.24 meters (50 ft.) until the shoreward edge of the shoal grass bed was encountered. Estimates of Cuban shoal grass, turtle grass, widgeon grass, and macroalgae cover were used to select donor sites that maximized Cuban shoal grass cover and shoot density, while minimizing secondary impacts to other seagrass species. Detailed survey results are provided in Appendix 2. Of the nine sites surveyed, all were viable for use as donor sites. However, due to the presence of archeological resources as well as stakeholder concerns, seven sites were selected and permitted for future transplanting activities.

Based on these findings in April 2019, the FWC began a 16-month pilot study to assess transplanting method and planting unit (PU) survival in a small 0.24-acre portion of the project area (Figure 2). The work was requested by the Florida Department of Environmental Protection and the United States Army Corps of Engineers during pre-application meetings to demonstrate transplant success prior to authorizing more expansive seagrass restoration within the project area. This pilot study was specifically designed to look at the effectiveness of transplanting Cuban shoal grass using bare-root methods into gridded arrays along the western shore of West Bay, St. Andrew Bay. Similar methods have proven effective in south Florida but have not been rigorously tested in temperate Florida and there was uncertainty regarding how well the method would work in a seasonal system exposed to winter senescence and storm-related disturbance.

In addition to testing the transplantation method, the pilot study reported herein also evaluated mitigation to two common threats in West Bay: bioturbation and sediment movement. Wooden dowels were used as a means to deter feeding and settlement by stingrays, primarily, which frequent the West Bay system and are known to cause harmful bioturbation effects to transplanted seagrass (Dr. Linda Fitzhugh, personal communication). Sediment movement has also been reported as a major roadblock in seagrass recovery in West Bay (https://myfwc.com/media/14173/carlson-roadblocks-panama-city.pdf). Altered bathymetry, unstable banks and the passage of large vessels within the GIWW appear to release sediments into the West Bay system (Brim & Handley 2007). Directions of seasonal winds and winter storms also contribute to sediment movement along the western shoreline of West Bay. All of

these are believed to be limiting factors in the natural recruitment of seagrasses to the restoration site.

The following describes the methods used, results found, and conclusions drawn from FWC's 2019-2020 seagrass transplantation pilot study at West Bay. Empirical information drawn from this work will be used to inform a larger seagrass transplantation effort currently under development by the FWC.

Methods

Seagrass Harvesting and Transplanting

A single donor site, located adjacent to Breakfast Point (Figure 4), was selected for use during FWC's 2019-2020 seagrass transplantation pilot study (see Appendix 2: donor site 7). The donor site was selected for its large size, adequate seagrass coverage, and close proximity to the recipient site (~4.5 km). FWC staff and volunteers harvested Cuban shoal grass over three days (April 16-18, 2019) using the bare root harvesting method. This method involved the removal of approx. 20-cm diameter units of seagrass with an approx. 20-cm (8-inch) shovel to a maximum depth of 8 centimeters. Sediments were rinsed from roots and rhizomes in situ and placed back into harvest depressions to minimize donor site recovery time. Harvesting units were spaced at a minimum of 2-meter intervals to avoid concentrated impacts to the donor site. In all cases, care was taken to avoid harvest of other species, particularly turtle grass; however, a small proportion (~10%) of widgeon grass was noted at time of collection. Plant material was stored in ambient seawater, shaded and immediately transported to the restoration site for PU production.

Planting units (PUs) were assembled by attaching at least, 3-5 rhizome sections, each with at least three Cuban shoal grass shoots and one apical meristem to a degradable metal sod staple, using paper and wire twist ties. Any remaining seagrass harvested not fitting the rhizome and apical meristem requirements were grouped in addition to those described above to increase the PUs thickness and avoid discarding seagrass. A total of 2,160 PUs were assembled and planted within five 0.40 hectare restoration plots (plots 2-6; Figure 2) on the same day seagrass was harvested. Plot locations were based on known light availability at light study stations 2, 3 and 4 where photosynthetically active radiation values ranged from 37.5% to 40.3% (Appendix 1). These levels are above the 33.3% light availability threshold reported as sufficient for Cuban shoal grass growth (Beal & Schmidt 2000). Each plot was separated into four planting clusters, six by eight meters each; Each cluster was divided into 12 planting sites, two meters square each (Figure 3). Nine PUs were placed within each planting site. Spacing between sites was approximately 2-m. Wooden pine dowels measuring 40.64 centimeters in length and 0.64 centimeters in diameter were inserted approx. 25 centimeters into the sediment at a frequency of 30 per m^2 within each site to inhibit stingray bioturbation (Figure 3).

Monitoring Protocol

Donor Sites

At the time of donor plant harvesting (April 16-18, 2019), thirty fixed quarter-meter square (0.5 m x 0.5 m) monitoring sites were haphazardly selected for tracking donor site recovery (Figure 4). Donor monitoring sites were marked with two approx. 1 meter polyvinyl

chloride (PVC) poles inserted into the sediment to a depth of 0.5 meter at opposite corners allowing the monitoring quadrat to be placed over the PVC poles in the same location and orientation during each monitoring event. Thirty non-fixed donor reference sites were then haphazardly selected within 10 meters of each fixed donor site at each monitoring event. Monitoring surveys were conducted over a one-year period at approximately 2-3 weeks, 6 weeks, 3 months, 6 months, 9 months, and 12 months post harvest.

Donor site surveys included recording the shoot density and percent frequency of occurrence for all seagrass species at each monitoring and reference site. Due to difficulties in identifying widgeon grass among the Cuban shoal grass-dominated seagrass population while snorkeling, shoot density and percent frequency data were combined under Cuban shoal grass. Quadrats were divided into 100 cells (5 cm x 5 cm each) and seagrass percent frequency determined by recording the presence/absence of each seagrass species (Cuban shoal grass/widgeon grass combined and turtle grass) in each cell. Five cells within each quadrat were randomly selected for shoot counts and fixed for the duration of the study.

Photographic data (overhead and oblique photographs) were taken at each monitoring donor site using a GoPro HERO7 Silver camera. Sediment depth was measured by recording the depth of the hole resulting from seagrass harvest relative to the substrate surface using a stiff rod placed horizontally along the sediment surface.

Recipient Sites

Three fixed 1-m^2 (1 m x 1 m) monitoring sites were haphazardly selected at each of the 20 planting clusters for a total of 60 recipient monitoring sites. Two approx.. 1 meter PVC poles were inserted into the sediment to a depth of 0.5 meters at opposite corners allowing the monitoring quadrat to be placed over the PVC poles in the same location and orientation during each survey event. Sixty non-fixed recipient reference sites were then haphazardly selected within 10 meters of each fixed recipient site at each monitoring event. Monitoring surveys were conducted over a 16-month period with the first year at the same frequency and time as donor site monitoring.

Recipient site surveys included estimates of shoot density and percent frequency of all seagrass species at each recipient monitoring and recipient reference site following a similar protocol as described above for donor sites above. Quadrats were divided into 100 cells (10 cm x 10 cm each) and seagrass percent frequency was determined by recording the presence/absence for transplanted Cuban shoal grass in each cell. Ten cells within each quadrat were randomly selected and fixed for the duration of the study to monitor Cuban shoal grass shoot counts.

Photographic data (overhead and oblique photographs) were taken at each monitoring recipient site using a GoPro HERO7 Silver camera. Sediment elevation was monitored at each of the 20 planting clusters by installing one 1.22-meter rebar stake 0.61 meter into the sediment on the east side of each planting group. At the time of rebar installation, the sediment surface height was marked with a fixed plastic cable tie and used as a reference to monitor changes in sediment elevation over the course of one year. At each monitoring event, a ruler was used to record any accretion (positive) or erosion (negative) of sediment by measuring the distance from the sediment surface to the cable tie.

Statistical Methods

Modeling objectives

Our general modeling objectives were threefold and focused on assessing spatial and temporal differences in (1) shoot count density at donor and recipient locations; (2) percent frequency of seagrass at donor and recipient locations; and (3) sediment elevation at donor and recipient locations. For donor locations, we fit separate models for turtle grass and Cuban shoal grass/widgeon grass for objectives (1) and (2), whereas for recipient locations we targeted Cuban shoal grass for transplantation. We used negative binomial regression to model shoot count density, logistic regression to model percent frequency of seagrass, and locally estimated scatterplot smoothing (loess) to model sediment elevation. All models were fitted in R v3.6.3 (R Core Team, 2020) using the glmmTMB package (negative binomial and logistic regression; Brooks et al. 2017) and ggplot2 (loess; Wickham 2016). Goodness-of-fit for all negative binomial and logistic regression models was assessed using a simulation-based approach for residual diagnostics, implemented in the R package DHARMa (Hartig 2019).

Model fitting and model selection

Shoot count and percent frequency modeling: donor sites

For both Cuban shoal grass/widgeon grass and turtle grass at donor locations, we fit six candidate shoot count (negative binomial) and percent frequency (logistic regression) models, with each model representing a different combination of (1) NumDays, or the number of days since seagrass was removed from the plot (here, days since April 16, 2019); (2) $NumDays^2$ (a quadratic relationship); (3) $NumDays^3$ (a cubic relationship); and (4) Site Type, a categorical variable representing either Monitoring or Reference plots. For three of the six candidate models, we included at least one predictor associated with NumDays along with the additive effect of Site Type. For the remaining three models, we replaced additive effects of Site Type with interactions between NumDays, $NumDays^2$, and $NumDays^3$ and Site Type. Additionally, to account for possible dependence among spatially-clustered groups of plots, we included a random intercept associated with each of the six spatially-clustered groups of plots (Figure 4; Gelman and Hill 2007); we note, however, that for turtle grass we excluded a single group as none of its sites ever supported the species. Following model-fitting, we again ranked each of the six candidate models and identified the most plausible combination of predictors by calculating Akaike's Information Criterion (Akaike 1973) with a small sample bias adjustment (AICc; Hurvich and Tsai 1989), and relative AICc weights (Burnham and Anderson 2002). We based all inferences on the best-approximating models.

Shoot count and percent frequency modeling: recipient sites

Similar to donor sites, for plots at recipient locations, we fit six candidate shoot count (negative binomial) and percent frequency (logistic regression) models, with each model representing a different combination of (1) NumDays, or the number of days since seagrass was translocated to the plot (here, days since April 16, 2019); (2) $NumDays^2$ (a quadratic relationship); (3) $NumDays^3$ (a cubic relationship); and (4) Plot Number, a categorical variable

representing the identity of each of the 5 spatially-clustered groups of survey plots. We note that for the recipient site shoot count models, we excluded cells that never supported seagrass, as our primary focus was to assess spatial and temporal differences given the cell supported seagrass at some point in time. For three of the six candidate models, we included at least one predictor associated with NumDays along with the additive effect of Plot Number, whereas the remaining three models replaced the additive effects of Plot Number with interactions between NumDays, $NumDays^2$, and $NumDays^3$ and Plot Number. We ranked each of the six candidate models as described for donor sites, and we based all inferences on the best-approximating models.

Sediment elevation modeling: donor and recipient sites

Due to limitations that prevented fitting linear regression models in a manner similar to shoot counts and percent frequency, we instead opted for a non-parametric loess approach to visually assess temporal changes in sediment elevation at (1) the recipient site, separately for each plot, and (2) at the donor site, averaged across groups. For both loess models we used a span of 0.8, where span ranges from 0 to 1 and represents the proportion of observations over which weighted running averages are calculated.

Results

Goodness-of-fit: shoot count and percent frequency modeling

For all shoot count and percent frequency models, the simulation-based assessments of model residuals indicated that all candidate negative binomial and logistic regression models provided adequate fit to the data.

Donor sites Cuban shoal grass/widgeon grass: shoot count modeling

The best approximating shoot count model for Cuban shoal grass/widgeon grass had 78% of the weight among candidate models and was 3.54 times more plausible than the next-best approximating model (Table 1). Parameter estimates (Table 2) and model predictions indicated that although monitoring plots generally had a lower mean shoot counts for the majority of the study period, there was no distinguishable difference in mean shoot count between monitoring and reference sites at the end of the study period (Figure 5).

Donor sites Cuban shoal grass/widgeon grass: percent frequency modeling

The best approximating percent frequency model for Cuban shoal grass/widgeon grass had 54% of the weight among candidate models and was 1.22 times more plausible than the next-best approximating model (Table 1). Parameter estimates (Table 2) and model predictions indicated that, similar to mean shoot count, although monitoring plots generally had lower percent frequency for the majority of the study period, there was no distinguishable difference in percent frequency between monitoring and reference sites at the end of the study period (Figure 6).

Donor sites turtle grass: shoot count modeling

The best approximating shoot count model for turtle grass had 90% of the weight among candidate models and was 9.41 times more plausible than the next-best approximating model

(Table 3). Parameter estimates (Table 4) and model predictions indicated that mean shoot count was similar between monitoring and reference plots throughout the study period (Figure 7).

Donor sites turtle grass: percent frequency modeling

The best approximating percent turtle grass frequency model had 100% of the weight among candidate models (Table 3). Parameter estimates (Table 4) and model predictions indicated that turtle grass percent frequency was similar between monitoring and reference plots throughout the study period (Figure 8).

Donor sites: sediment recovery

For the donor site, the loess model indicated that, combined across all 6 groups of survey plots, sediment elevation gradually increased toward 0 cm; hence depressions caused by removal of seagrass generally recovered over the course of the study period (Figure 9).

Recipient sites: shoot count modeling

The best approximating percent frequency model for recipient sites had 94% of the weight among candidate models (Table 5). Parameter estimates (Table 6) and model predictions indicated that temporal changes in mean shoot count varied among plot, with plots 2, 3, and 4 exhibiting, on average, higher shoot counts than plots 5 and 6 (Figure 10).

Recipient sites: percent frequency modeling

The best approximating percent frequency model for recipient sites had 100% of the weight among candidate models (Table 5). Parameter estimates (Table 7) and model predictions indicated that, similar to mean shoot count, temporal changes in percent frequency varied among plots, with plots 2, 3, and 4 exhibiting, on average, higher percent frequency than plots 5 and 6 throughout the study period (Figure 11).

Recipient sites: sedimentation

For the recipient site, the loess model indicated that sediment elevation generally increased throughout the study period, although the rate of increase varied among plots; specifically, there appeared to be a more rapid increase in sediment elevation at plots 5 and 6, particularly within the first 100 days following transplantation (Figure 12).

Discussion

Donor Sites

Cuban shoal grass/widgeon grass shoot density & percent frequency

Immediately (15 days) following the harvest of transplanted material from donor areas, the estimated difference in mean Cuban shoal grass/widgeon grass shoot density at monitoring sites vs. reference sites was 1.80 shoots per cell (Figure 5). Over the course of the 388-day monitoring period, this difference declined to 0.61 shoots per cell indicating recovery towards

pre-impact conditions. While mean Cuban shoal grass/widgeon grass percent frequency did not immediately diverge among harvested and non-harvested reference sites, a maximum difference of 14% was recorded on day 242. This was followed by recovery of mean Cuban shoal grass/widgeon grass percent frequency to values equivalent to and then exceeding reference sites (Figure 6). Higher Cuban shoal grass percent frequency values at monitoring sites compared to reference sites are most likely explained by the occasional inclusion of bare patches, prevalent through the entire donor area, within haphazardly selected reference quadrats towards the end of the study period.

Turtle grass shoot density & percent frequency

Locations containing turtle grass were avoided during harvesting efforts, thus our monitoring sites contained little to no turtle grass frequency (Figures 7 & 8). Recovery rates are also much slower for turtle grass than Cuban shoal grass, and so it was expected that monitoring sites would not mimic patterns of the reference sites during our pilot study (Fonseca et al. 1994). However, by day 250 post harvest, mean turtle grass shoot density and percent frequency values had approached reference conditions (Figures 7 & 8). These results show harvesting techniques did not result in lasting damage to the donor beds, and that even the slowest growing of the impacted species (turtle grass) was able to recover in terms of shoots density and areal cover in as little as 12-months post impact.

Donor site sediment recovery

Mean donor site sediment depth decreased from 4.38 cm on day 15 to 0.60 cm on day 388, indicating that substrates disturbed during harvesting of plant material continued to infill for the duration of the 12-month study (Figure 9). At some donor site locations, species such as Gulf toadfish (*Opsanus beta*) and blue crab (*Callinectes sapidus*) were observed utilizing harvest depressions as well as those areas directly adjacent to PVC marker poles. The additional bioturbation likely affected our measurements of sediment height and so all time-points for sites with toadfish or blue crab residents were excluded from statistical analyses.

Recipient Sites

Cuban shoal grass shoot density

Mean Cuban shoal grass shoot density at recipient sites remained steady over the course of the pilot study indicating continued survival of transplanted shoots into the second (first full) growing season (Figure 10). Following the first 100 days, shoot density increased or remained relatively stable at plots 2, 3 and 4 and decreased at plots 5 and 6. The design of PUs consisted of multiple runners, each with several shoots. Attrition of damaged shoots and loss of expanding runners during the first 100 days likely contributed to higher variance recorded during this time (Figure 10). Also, due to the centralized installation pattern (i.e., nine PUs placed within the center of each site; Figure 3), some randomly selected cells within the 1-m^2 monitoring quadrat were not located over the central cluster of PUs. While this randomized cell sampling scheme was useful for tracking seagrass expansion, it may not have accurately captured shoot density during the initial phase of establishment. Finally, small translational errors in quadrat placement during successive monitoring events may have affected our estimates of shoot density. For these

reasons, shoot counts were less consistent across early monitoring periods than originally intended. To account for this in our statistical analysis, cells that never contained seagrass were excluded.

Cuban shoal grass percent frequency

Although an initial estimate of percent frequency was not obtained at the time of transplanting, we targeted 10% seagrass frequency based on PU size and arrangement within each site (Figure 3). From day 20 to day 482, mean Cuban shoal grass percent frequency at recipient sites increased from 22% to 37% across all plots (Figure 11) suggesting that our installation target had been met. The increase was largely driven by plots 2, 3, and 4, where average shoal grass percent frequency grew from 21%, 24%, and 24% respectively (day 20) to 65%, 40%, and 60% respectively (day 482). In contrast, shoal grass percent frequency decreased at plots 5 and 6 from 25% and 28% respectively (day 20) to 12% and 8% respectively (day 482). As with shoot density, small translational errors described above could have resulted in the bisecting of single PUs by two or more grid cells or an overestimation of shoot presence within individual cell boundaries made in turbid water conditions.

Of the 60 sites monitored, 57 (95%) contained at least 1% seagrass frequency one-year post transplantation. This can be compared to other studies using the bare-root transplantation method. For example, in a study including sites located in North Carolina and Florida, Fonseca et al. (1994) observed 89% PU survival at day 96 followed by a slight decline to 85% PU survival at 469 days. In the present study, at approximately 100 days, plots 2-4 showed a positive trend for expansion compared to the decreasing trend observed at plots 5 and 6 (Figure 11). Cuban shoal grass percent frequency followed seasonal variation and had high success (~89%) for monitoring sites expanding to greater than 15% frequency at plots 2, 3, and 4. Plots 5 and 6 had the lowest survival rates with nearly 71% of monitoring sites with frequency values near or below 10% on day 482. Temporal trends in areal frequency and shoot density were similar at the recipient sites (Figure 10). Together, they suggest that the area within and surrounding plots 2-4 is most suitable for seagrass restoration success under current site conditions.

Recipient site sedimentation

Previous research in the project area indicates Cuban shoal grass growth possible up to a mean depth of 0.91 ± 0.2 m SD (Fitzhugh 2012). Our data were consistent with this, as our most successful plots, 2 and 4, were planted along the 1-meter isobath (Figure 2). They had the greatest seagrass expansion and shoot density (Figures 10 & 11) while incurring the least amount of net sediment movement (Figure 12) over the course of the study.

Sediment plays a significant role in seagrass survival. As indicated by Fonseca et al. (1998), when 25% of seagrass shoots were buried, only 75% of the plants survived. This is in contrast with a 5% survival rate when 75% of seagrass shoots were covered by sediments. As shown in Figure 12, at approximately 100 days post transplantation, plots 5 and 6 had higher average sediment accumulation values of 3.95 cm and 4.42 cm, respectively, and the lowest average percent frequency and shoot density (Figure 10 & 11). The opposite was true for plots 2-4, which had the lowest average sediment accumulation values of 2.79 cm, 2.35 cm, 1.51 cm, respectively (Figure 12) and highest percent frequency and shoot density (Figure 10 & 11).

While the FWC light availability study (Appendix 1) found that all transplanted plots had suitable light levels for seagrass survival, it was expected that the shoreward sites would be most suitable for seagrass transplantation due to their higher light levels and natural seagrass expansion at this depth adjacent to and north of the pilot study area. However, the percent frequency and shoot count success of plot 2 and 4 indicate there may be a tradeoff between light availability and wave, tidal, and wind energy affecting sediment movement closer to shore at the project site.

Our initial estimates of sediment movement support others' claims that high sediment transport is a major roadblock to seagrass recovery in West Bay (https://myfwc.com/research/habitat/seagrasses/projects/roadblocks/); Sediment movement at West Bay is certainly of a magnitude that can affect Cuban shoal grass growth and survival. This is supported by field observations of naturally recruiting seagrass patches in the vicinity of the pilot study plots that showed mounding of sediments. These areas were likely inhibited by the same activation inhibition dynamics governing seagrass expansion in other high energy environments. A more robust sediment monitoring effort will be needed to better understand the interaction of sediment movement and PU survival in the system. To minimize the confounding influences of above-sediment structure and to better capture the spatial pattern of sediment movement, we recommend an array of sub-surface plexiglass plates such as those used by Bell et. al (1999) as an alternate to the rebar and cable tie method used here.

Bioturbation

Although no interactions between stingrays and the bioturbation stakes were observed during the transplanting or monitoring efforts, stingrays were noted in the area, as were stingray feeding pits on multiple occasions. It is reasonable to assume that stingrays attempted to feed within our transplant site and we suspect that the wooden pine dowels were an effective barrier to the disturbance of the newly planted PUs. In practice, we estimate that the above-ground portion of the stakes remained intact for 5 months. Once degraded, the remaining below-ground portion of the stakes may have continued to limit stingray feeding. Stingray sediment disturbance has been a major cause of loss in previous attempts to transplant seagrass in West Bay (Linda Fitzhugh, personal communication) and has caused over 50% of PU loss in Tampa Bay (Fonseca et al. 1994). The use of stingray bioturbation exclusion stakes or dowel rods was previously tested in the southern portion of West Bay by Dr. Linda Fitzhugh (Gulf Coast State College, personal communication) and, as opposed to other methods such as exclusion cages, allows for adequate sunlight to reach transplanted seagrass. Exclusion cages were tested in West Bay in 2003-2004 and resulted in shading of seagrass and the promotion of heavy algae growth preventing seagrass restoration success (Lamb 2006). While effectively deterring stingrays from the project site, wooden dowels may also have the added benefit of stabilizing sediments, although the magnitude of this protection is unknown.

Ecosystem services

While tracking the presence and number of associated species was not included in the monitoring protocol for this pilot study, a number of juvenile finfish were frequently observed among the transplanted seagrasses. For example, an approximately 15-day old Atlantic spadefish (*Chaetodipterus faber*) was seen on May 30th and two approximately 60-day old juveniles were

seen on July 18th, 2019. Other species observed include juvenile blue crabs (*Callinectes sapidus*), gobies (Gobiidae spp.), arthropods (Malacostraca spp.), and pinfish (*Lagodon rhomboides*). These sightings were mostly made at plots 2-4, further indicating the suitability of the area for providing future ecosystem services including fishery enhancement (Beck et al. 2001; Costanza et al. 1997; Heck et al. 2003).

Conclusion

Study results indicate that donor sites recovered to baseline within one year of harvest and that bare root transplantation of Cuban shoal grass is a viable method for seagrass restoration at West Bay, a segment of St. Andrew Bay, Florida. Of the 60 sites planted with 9 PUs each, 95% of the surveyed sites monitored contained at least some seagrass at one year post transplantation. While significant PU expansion or coalescence would not be expected at this time, our monitoring results demonstrate that PUs installed in the spring, prior to the growing season, can recover from damage incurred by transplantation, establish and overwinter, maintaining the planting array into the second (i.e., first full) growing season.

While not dispositive, the observation of stingray feeding in the vicinity of the transplanting site and the lack of associated damage to the PUs indicates that the effort to curb bioturbation using wooden stakes was successful and may have continued even after the above-ground portion of the pine dowels eroded (~5 months). Further, it is reasonable to assume that the regular grid of stakes afforded the newly transplanted PUs some measure of protection from sediment movement, helping to stabilize sediments, particularly during the first weeks after planting.

Our preliminary, coarse-grained efforts to document sediment movement confirmed that the western shore of West Bay is a high energy environment exposed to wind-waves and associated long-shore currents. The magnitude of net sediment movement appeared sufficient to affect seagrass establishment; however, our methods were unable to fully characterize the temporal dynamics of the disturbance regime or to map it effectively at the landscape scale. To remedy these limitations, we propose tracking changes in sediment elevation using a more finely grained array of subsurface sediment height plates following the methods of Bell et al. (1999). In the meantime, the area in and around plots 2 and 4 appear to be most ideal for further seagrass transplantation work in this system.

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FIGURES

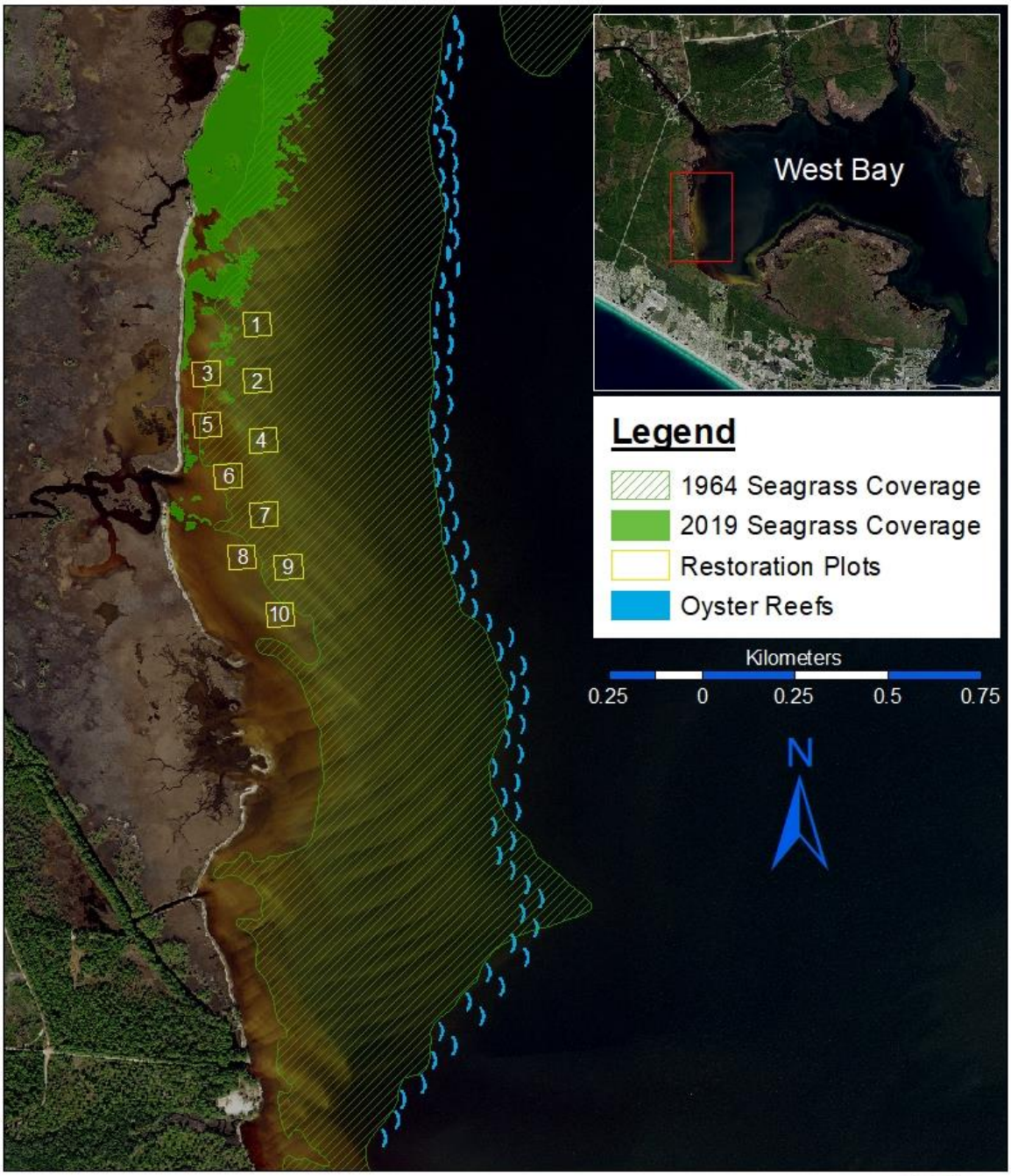


Figure 1: FWC's seagrass restoration site is located along the western shore of West Bay, St. Andrew Bay, where ten 0.40 hectare restoration plots were permitted for seagrass transplantation.

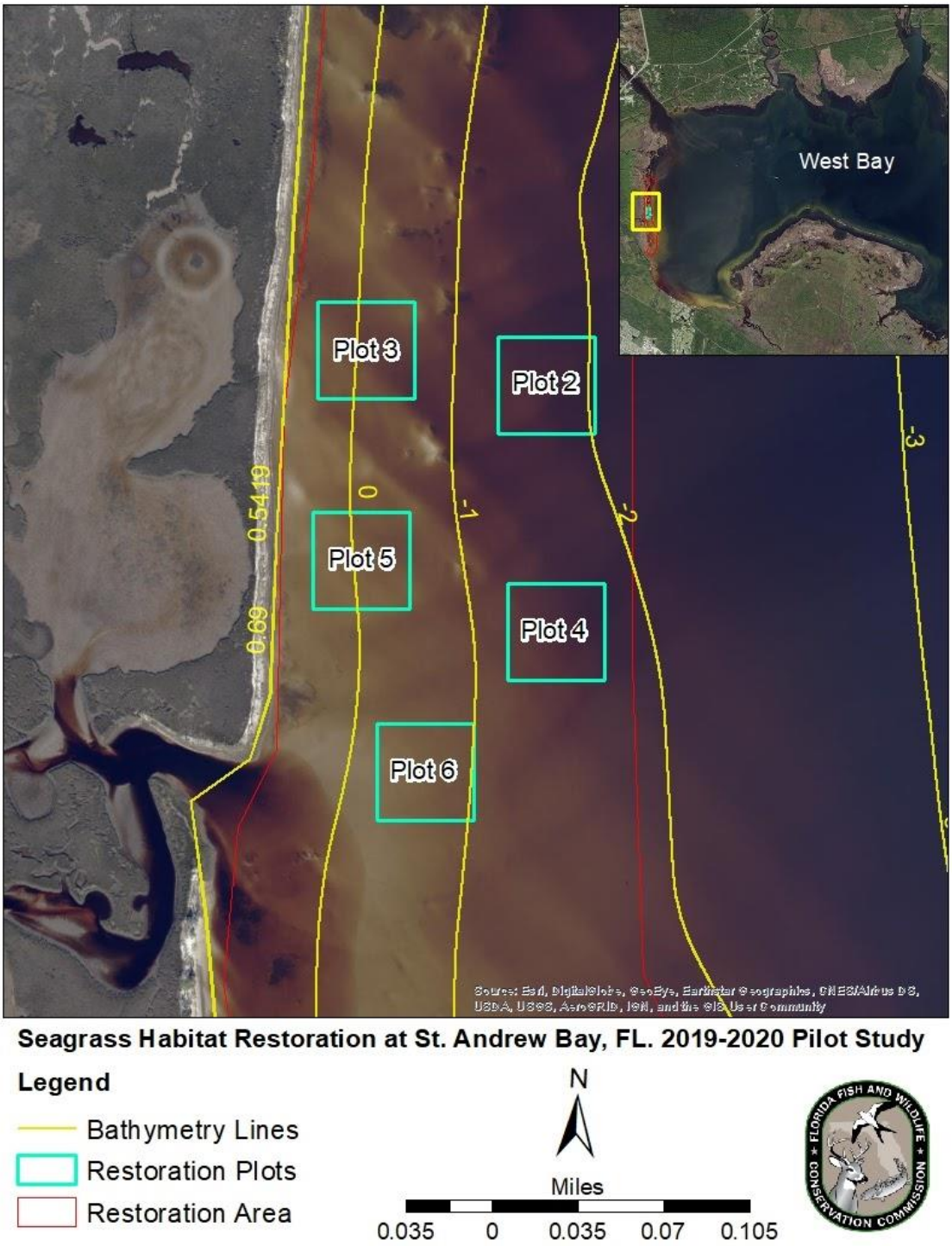


Figure 2: FWC's 2019-2020 seagrass transplantation pilot study evaluated five restoration plots (numbered 2-6) at varied distances from the shoreline and depths ranging from 0 to 2 feet below sea level (NAVD88).

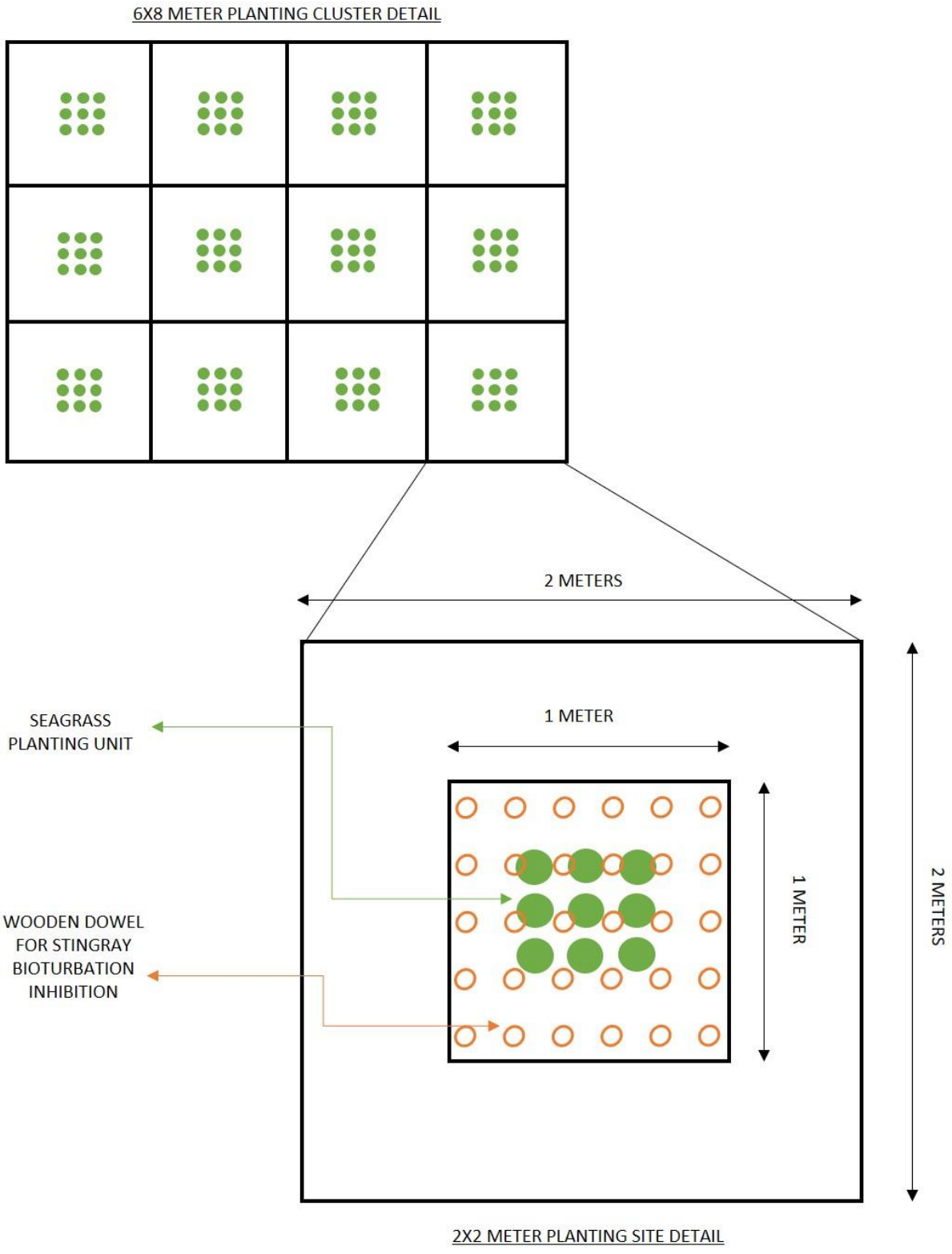


Figure 3: The detailed plan for transplanting seagrass at recipient sites involved the placement of nine centralized PUs within each of twelve 2 x 2 meter sites located within a single 8 x 6 meter planting cluster. A total of 20 planting clusters (4 each) were distributed across the 5 restoration plots.

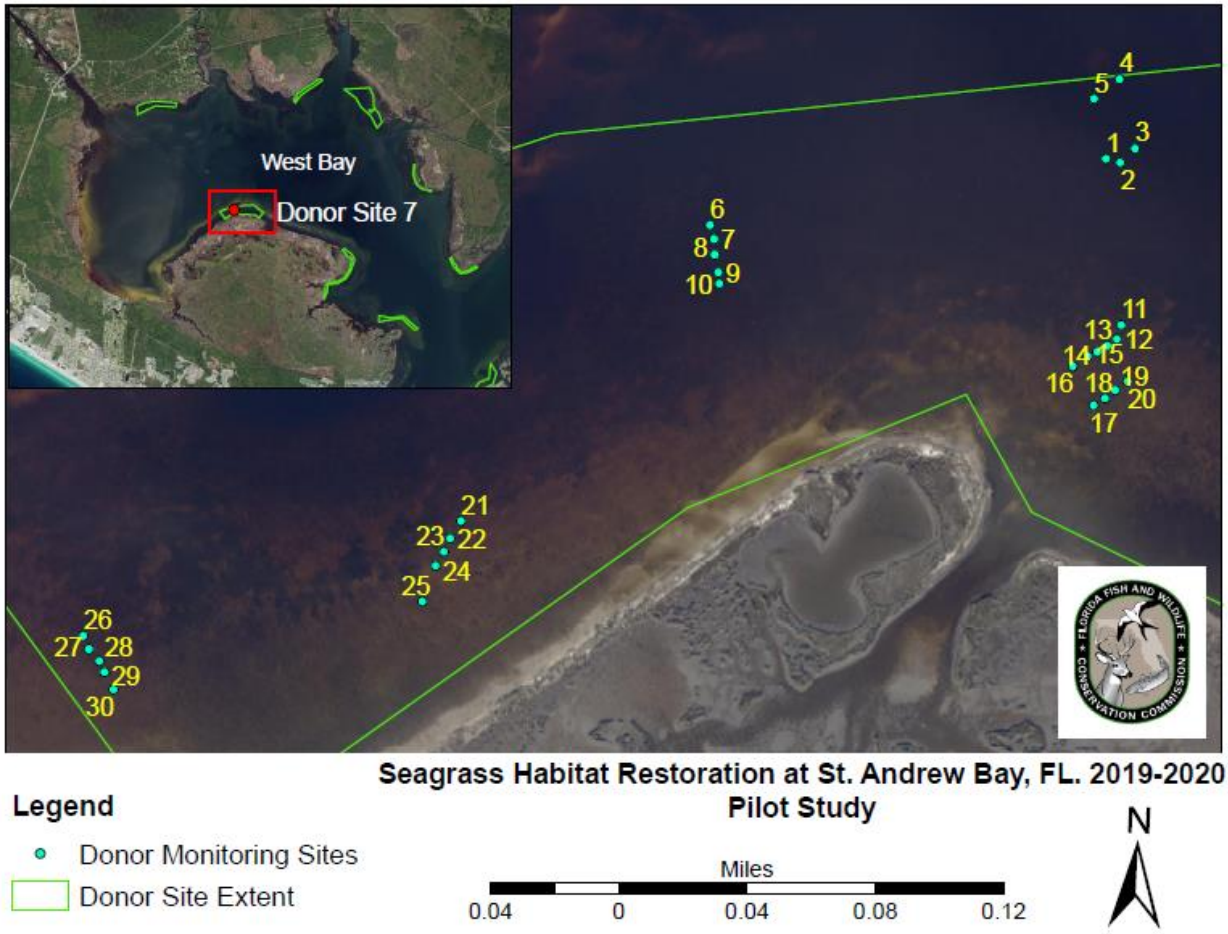


Figure 4: FWC's 2019-2020 seagrass transplantation pilot study donor site was located north of Breakfast Point and directly east of the project recipient site. Thirty fixed 0.5 m x 0.5 m monitoring sites were haphazardly selected for tracking donor site recovery. These were paired with 30 reference sites haphazardly selected at each monitoring event.

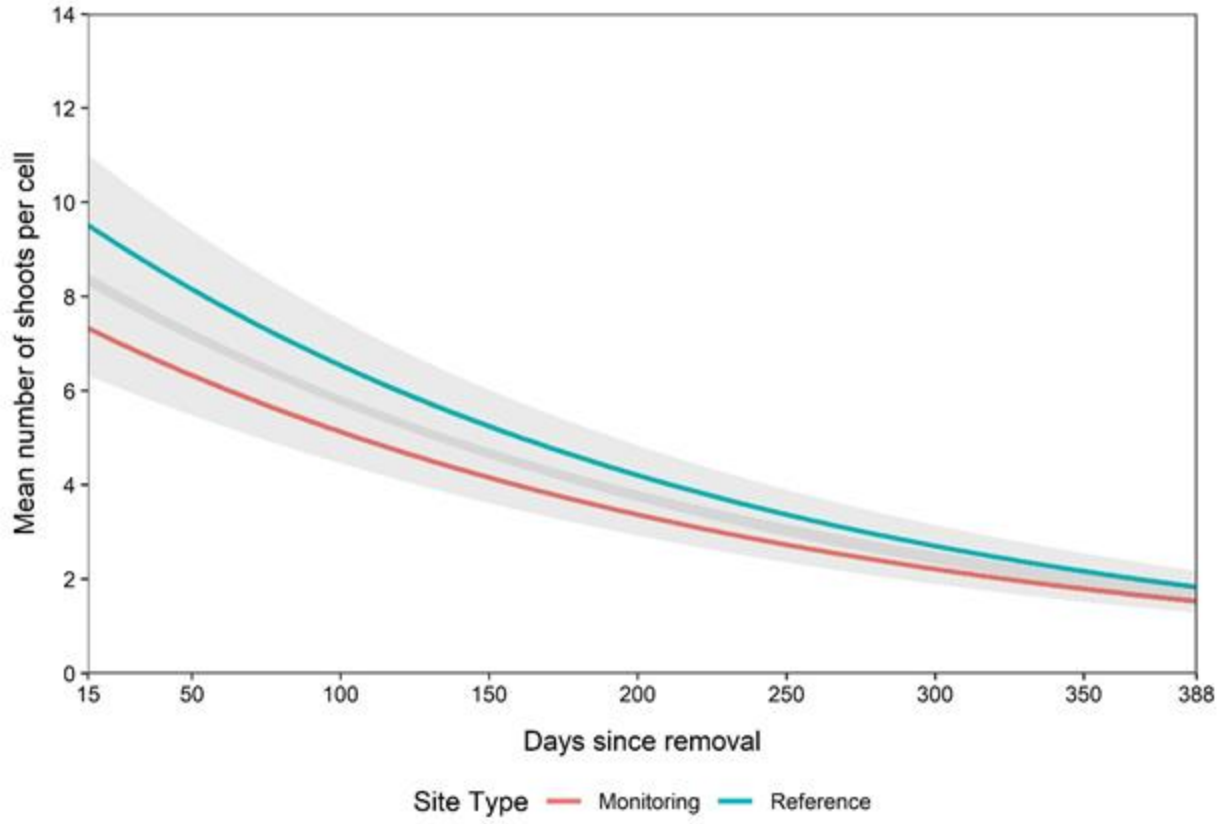


Figure 5: Predictions based on the best-approximating negative binomial regression model of mean Cuban shoal grass shoot density in relation to the number of days since removal for monitoring and reference plots at the donor site.

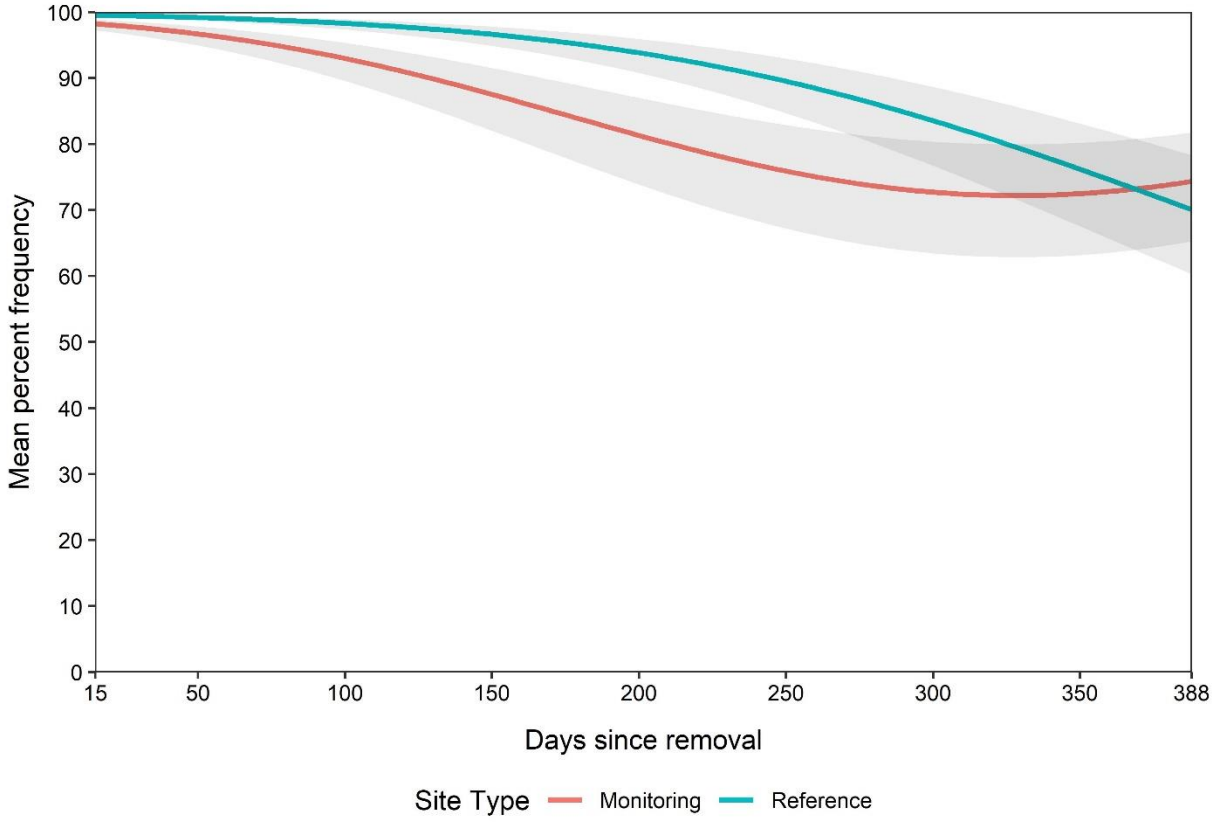


Figure 6: Predictions based on the best-approximating logistic regression model of mean Cuban shoal grass percent frequency in relation to the number of days since removal for monitoring and reference plots at the donor site.

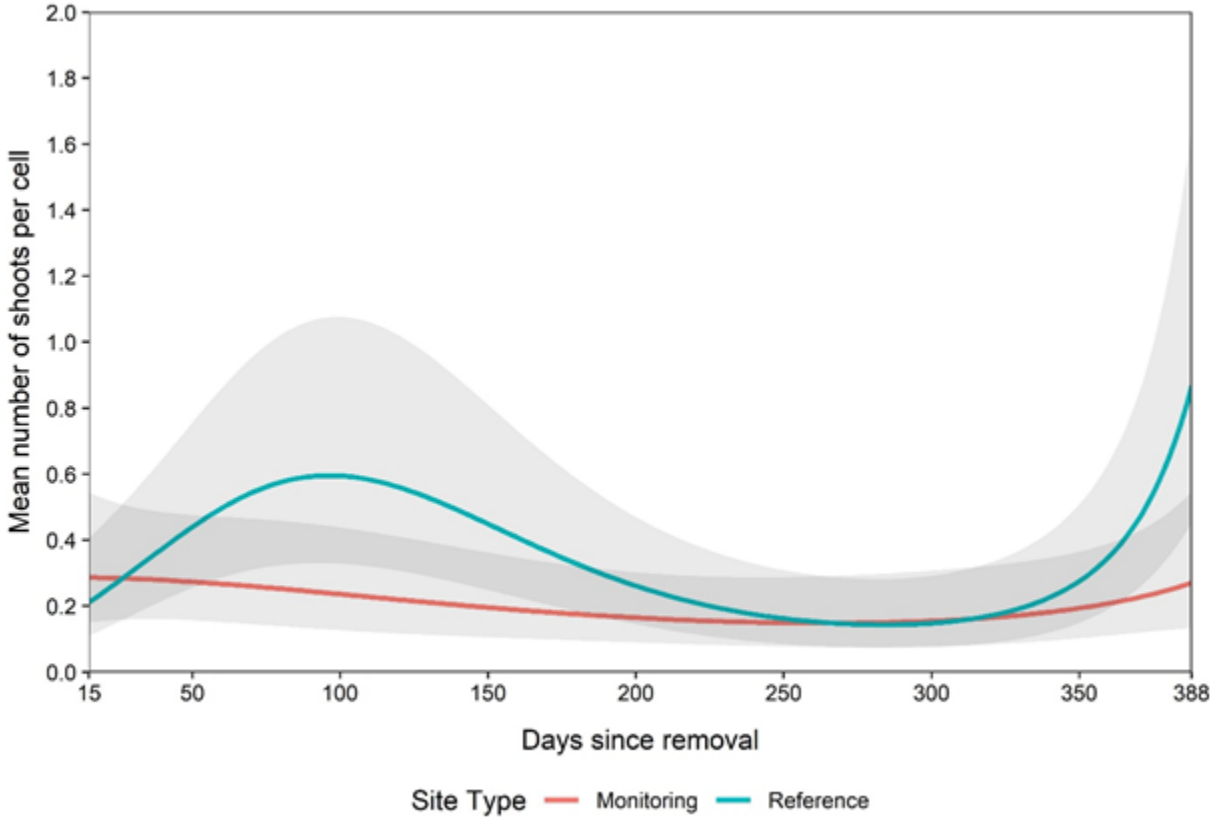


Figure 7: Predictions based on the best-approximating negative binomial regression model of mean turtle grass shoot count density in relation to the number of days since removal for monitoring and reference plots at the donor site.

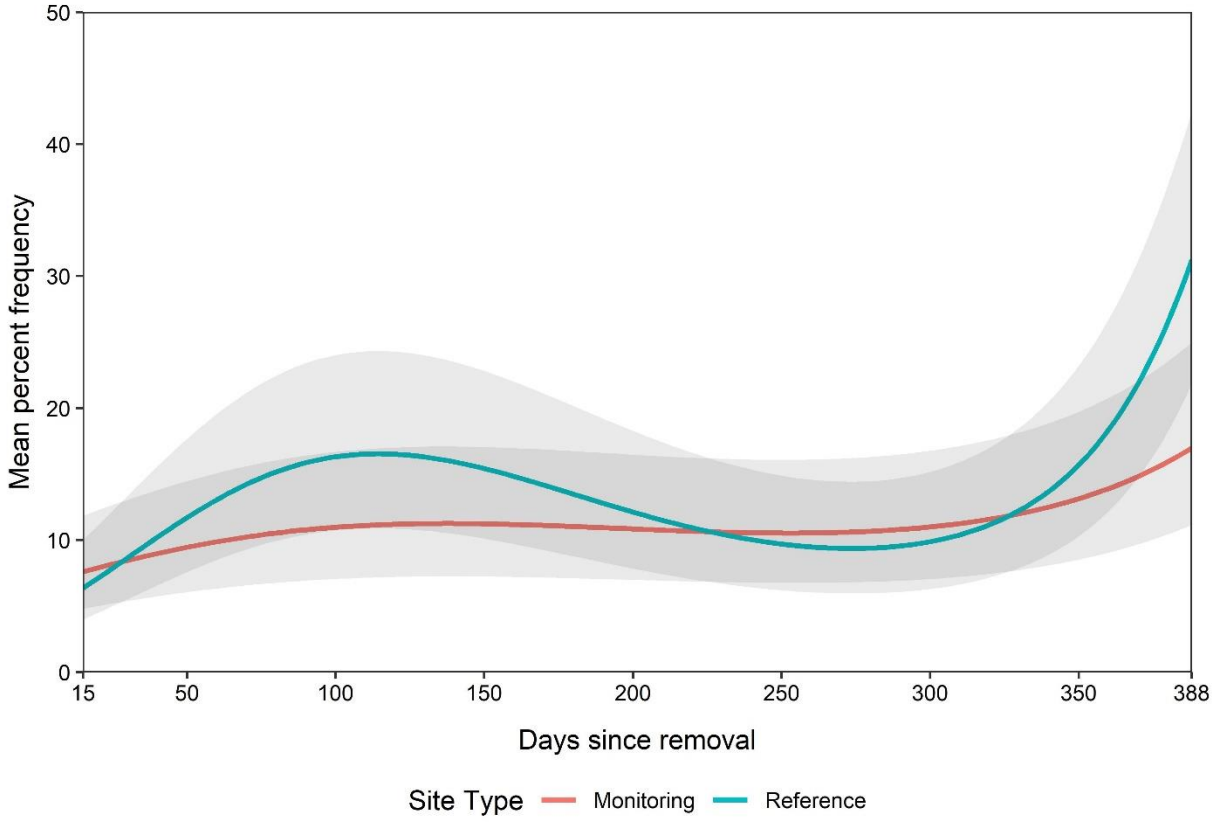


Figure 8: Predictions based on the best-approximating logistic regression model of mean turtle grass percent frequency in relation to the number of days since removal for monitoring and reference plots at the donor site.

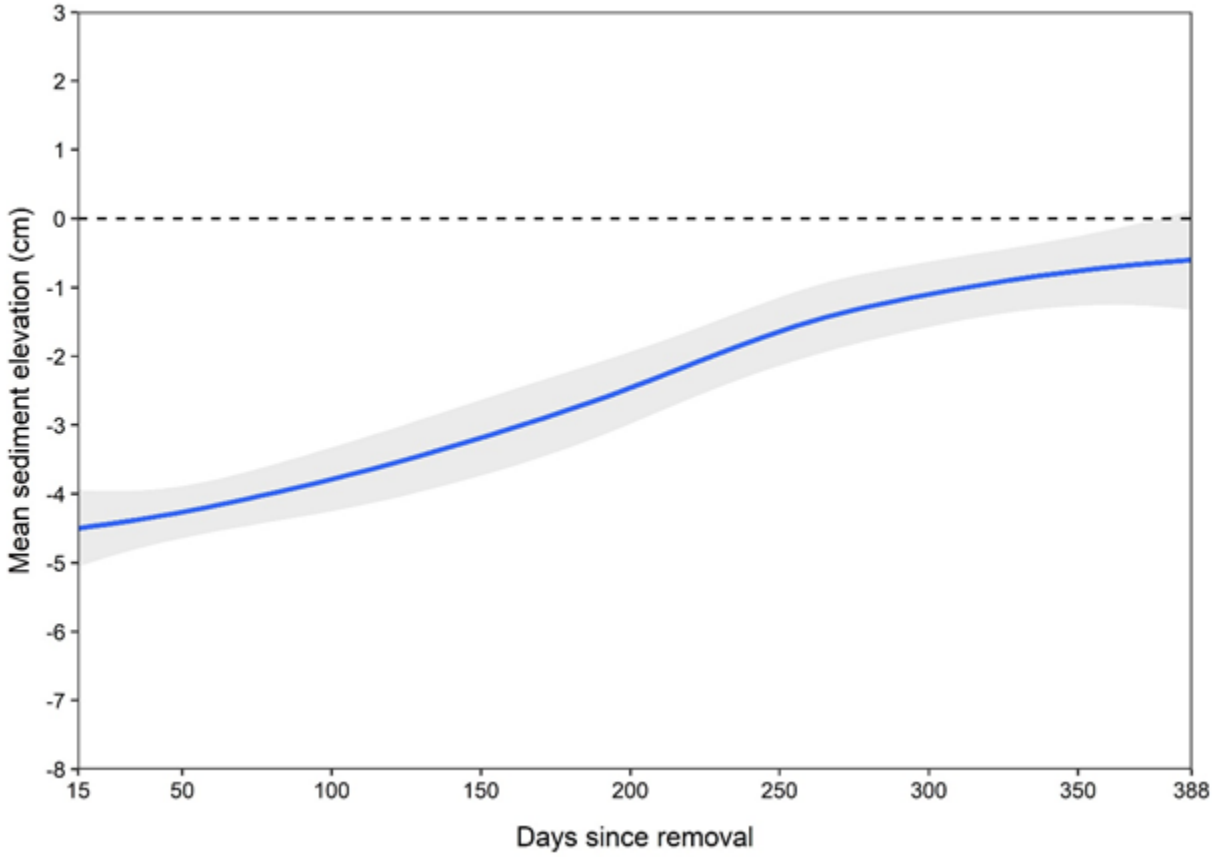


Figure 9: Donor sediment elevation based on loess model.

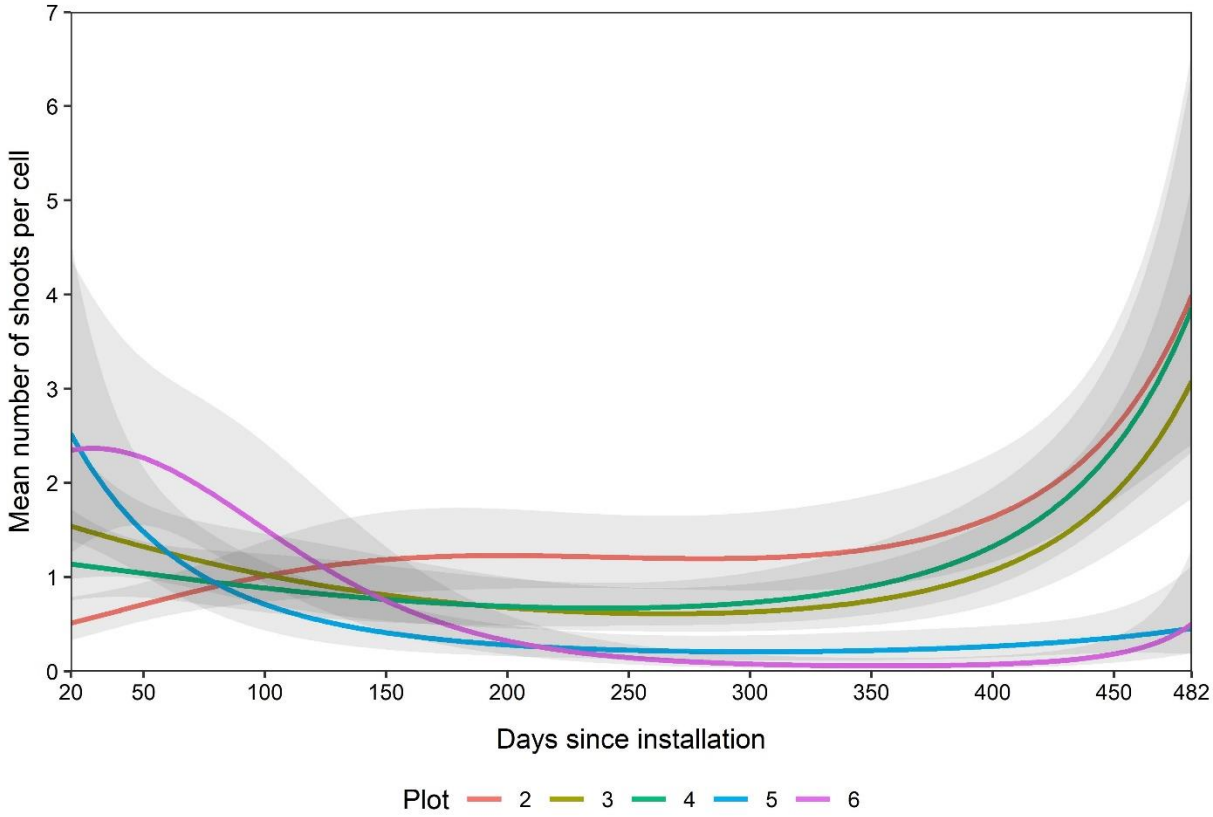


Figure 10: Predictions based on the best-approximating negative binomial regression model of Cuban shoal grass shoot density in relation to the number of days since installation for each of the five recipient site plots.

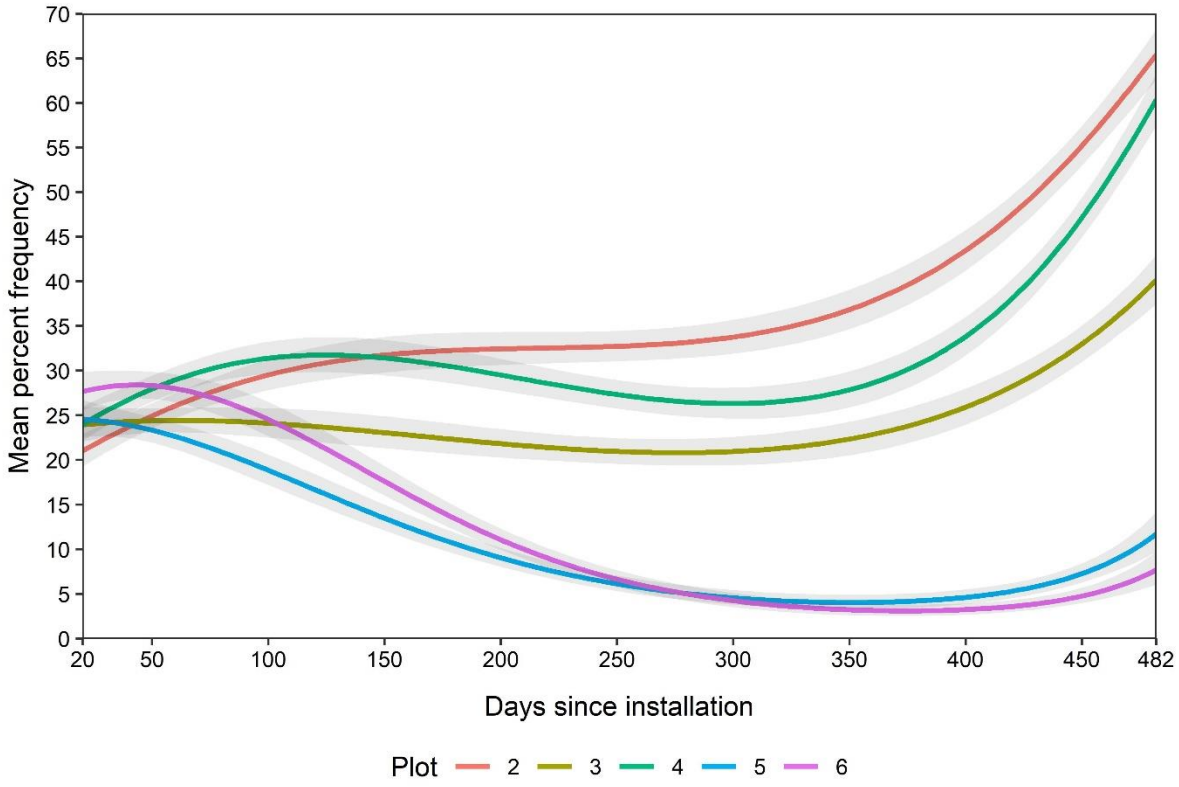


Figure 11: Predictions based on the best-approximating logistic regression model of mean Cuban shoal grass percent frequency in relation to the number of days since installation for each of the five recipient site plots.

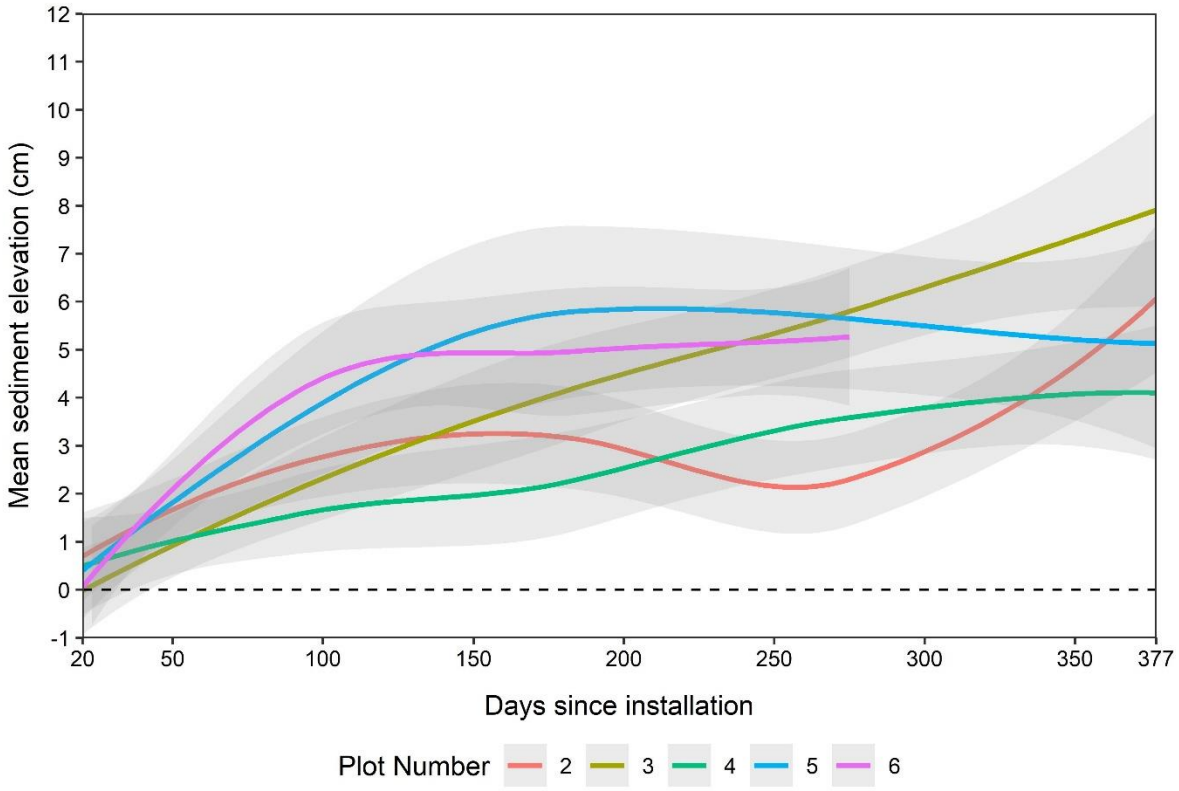


Figure 12: Recipient sediment elevation based on loess model.

TABLES

Table 1: Model parameters (model), number of parameters (K), AICc, ΔAICc, and Akaike weights (w) for the candidate sets of negative binomial and logistic regression models relating donor site Cuban shoal grass/widgeon grass shoot density and percent frequency to the number of days since removal (NumDays) and site type (monitoring vs reference).

Model	K	log likelihood	AICc	ΔAICc	weight
Cuban shoal grass/widgeon grass number of shoots ~ negative binomial regression					
NumDays + $NumDays^2$ + $NumDays^3$ + SiteType	7	-4448.57	8911.20	0.00	0.78
NumDays + $NumDays^2$ + $NumDays^3$ + SiteType + NumDays × SiteType + $NumDays^2$ × SiteType + $NumDays^3$ × SiteType	10	-4446.79	8913.70	2.51	0.22
NumDays + $NumDays^2$ + SiteType	6	-4474.03	8960.10	48.91	0.00
NumDays + $NumDays^2$ + SiteType + NumDays × SiteType + $NumDays^2$ × SiteType	8	-4472.75	8961.60	50.38	0.00
NumDays + SiteType	5	-4511.82	9033.70	122.49	0.00
NumDays + SiteType + NumDays × SiteType	6	-4511.56	9035.20	123.98	0.00
Cuban shoal grass/widgeon grass percent frequency ~ logistic regression					
NumDays + $NumDays^2$ + SiteType + NumDays × SiteType + $NumDays^2$ × SiteType	7	-10245.39	20504.80	0.00	0.55
NumDays + $NumDays^2$ + $NumDays^3$ + SiteType + NumDays × SiteType + $NumDays^2$ × SiteType + $NumDays^3$ × SiteType	9	-10243.58	20505.20	0.38	0.45
NumDays + $NumDays^2$ + SiteType	5	-10395.80	20801.60	296.83	0.00
NumDays + $NumDays^2$ + $NumDays^3$ + SiteType	6	-10395.66	20803.30	298.54	0.00
NumDays + SiteType + NumDays × SiteType	5	-10407.62	20825.20	320.47	0.00
NumDays + SiteType	4	-10503.34	21014.70	509.90	0.00

Table 2: Parameter estimates, standard errors (SE) and 95% confidence intervals (Lower and Upper) from the best-approximating negative binomial (shoot counts) and logistic regression (percent frequency) models relating donor site Cuban shoal grass/widgeon grass shoot counts and percent frequency to the number of days since installation (NumDays) and site type.

Parameter	Estimate	SE	Lower	Upper
Shoot count (log link)				
Fixed effects				
Intercept	1.179	0.076	1.030	1.329
SiteType	0.242	0.036	0.171	0.313
NumDays	-1.025	0.057	-1.136	-0.914
$NumDays^2$	0.043	0.035	-0.027	0.112
$NumDays^3$	0.253	0.035	0.184	0.322
Theta	2.887			
Random effect				
Intercept (Group)	0.162			
Percent frequency (logit link)				
Fixed effects				
Intercept	1.788	0.219	1.358	2.218
SiteType	1.372	0.061	1.252	1.491
NumDays	-1.336	0.040	-1.414	-1.258
$NumDays^2$	0.535	0.032	0.473	0.597
NumDays × SiteType	-0.360	0.081	-0.519	-0.201
$NumDays^2$ × SiteType	-0.337	0.056	-0.446	-0.227
Random effect				
Intercept (Group)	0.531			

Table 3: Model parameters (model), number of parameters (K), AICc, ΔAICc, and Akaike weights (w) for the candidate sets of negative binomial and logistic regression models relating donor site turtle grass shoot density and percent frequency to the number of days since removal (NumDays) and site type (monitoring vs reference).

Model	K	log likelihood	AICc	ΔAICc	weight
Turtle grass number of shoots ~ negative binomial regression					
NumDays + $NumDays^2$ + $NumDays^3$ + SiteType + NumDays × SiteType + $NumDays^2$ × SiteType + $NumDays^3$ × SiteType	10	-1058.63	2137.40	0.00	0.90
NumDays + $NumDays^2$ + $NumDays^3$ + SiteType	7	-1063.91	2141.90	4.49	0.10
NumDays + $NumDays^2$ + SiteType	6	-1072.10	2156.30	18.86	0.00
NumDays + $NumDays^2$ + SiteType + NumDays × SiteType + $NumDays^2$ × SiteType	8	-1070.91	2157.90	20.51	0.00
NumDays + SiteType + NumDays*SiteType	6	-1074.35	2160.70	23.35	0.00
NumDays + SiteType	5	-1075.97	2162.00	24.59	0.00
Turtle grass percent frequency ~ logistic regression					
NumDays + $NumDays^2$ + $NumDays^3$ + SiteType + NumDays × SiteType + $NumDays^2$ × SiteType + $NumDays^3$ × SiteType	9	-11342.55	22703.10	0.00	1.00
NumDays + $NumDays^2$ + $NumDays^3$ + SiteType	6	-11387.78	22787.60	84.47	0.00
NumDays + $NumDays^2$ + SiteType + NumDays × SiteType + $NumDays^2$ × SiteType	7	-11476.55	22967.10	264.00	0.00
NumDays + $NumDays^2$ + SiteType	5	-11495.02	23000.10	296.96	0.00
NumDays + SiteType + NumDays × SiteType	5	-11496.79	23003.60	300.49	0.00
NumDays + SiteType	4	-11510.67	23029.30	326.25	0.00

Table 4: Parameter estimates, standard errors (SE) and 95% confidence intervals (Lower and Upper) from the best-approximating negative binomial (shoot counts) and logistic regression (percent frequency) models relating donor site turtle grass shoot counts and percent frequency to the number of days since installation (NumDays) and site type.

Parameter	Estimate	SE	Lower	Upper
Shoot count (log link)				
Fixed effects				
Intercept	-1.688	0.310	-2.296	-1.079
SiteType	0.735	0.265	0.216	1.254
NumDays	-0.477	0.319	-1.101	0.148
$NumDays^2$	0.117	0.217	-0.308	0.542
$NumDays^3$	0.174	0.204	-0.225	0.573
NumDays × SiteType	-0.912	0.443	-1.781	-0.043
$NumDays^2$ × SiteType	-0.675	0.304	-1.270	-0.079
$NumDays^3$ × SiteType	0.803	0.287	0.240	1.365
Theta	0.248			
Random effect				
Intercept (Group)	0.537			
Percent frequency (logit link)				
Fixed effects				
Intercept	-2.075	0.247	-2.559	-1.592
SiteType	0.300	0.062	0.178	0.422
NumDays	-0.093	0.074	-0.238	0.052
$NumDays^2$	-0.151	0.054	-0.257	-0.045
$NumDays^3$	0.221	0.049	0.126	0.316
NumDays × SiteType	-0.605	0.104	-0.809	-0.402
$NumDays^2$ × SiteType	-0.328	0.076	-0.476	-0.180
$NumDays^3$ × SiteType	0.505	0.068	0.372	0.638
Random effect				
Intercept (Group)	0.542			

Table 5: Model parameters (model), number of parameters (K), AICc, ΔAICc, and Akaike weights (w) for the candidate sets of negative binomial and logistic regression models relating recipient site shoot density and percent frequency to the number of days since removal (NumDays) and plot number.

Model	K	log likelihood	AICc	ΔAICc	weight
Shoot counts ~ negative binomial regression					
$NumDays + NumDays^2 + NumDays^3 + SiteType + NumDays \times SiteType + NumDays^2 \times SiteType + NumDays^3 \times SiteType$	21	-3139.30	6321.00	0.00	0.94
$NumDays + NumDays^2 + SiteType + NumDays \times SiteType + NumDays^2 \times SiteType$	16	-3147.12	6326.50	5.49	0.06
$NumDays + SiteType + NumDays \times SiteType$	11	-3178.32	6378.70	57.78	0.00
$NumDays + NumDays^2 + NumDays^3 + SiteType$	9	-3199.66	6417.40	96.44	0.00
$NumDays + NumDays^2 + SiteType$	8	-3202.62	6421.30	100.34	0.00
$NumDays + SiteType$	7	-3233.33	6480.70	159.74	0.00
Percent Frequency ~ logistic regression					
$NumDays + NumDays^2 + NumDays^3 + SiteType + NumDays \times SiteType + NumDays^2 \times SiteType + NumDays^3 \times SiteType$	20	-21339.52	42719.10	0.00	1.00
$NumDays + NumDays^2 + SiteType + NumDays \times SiteType + NumDays^2 \times SiteType$	15	-21467.08	42964.20	245.11	0.00
$NumDays + SiteType + NumDays \times SiteType$	10	-21622.52	43265.00	545.99	0.00
$NumDays + NumDays^2 + NumDays^3 + SiteType$	8	-22193.25	44402.50	1683.44	0.00
$NumDays + NumDays^2 + SiteType$	7	-22288.04	44590.10	1871.03	0.00
$NumDays + SiteType$	6	-22507.89	45027.80	2308.74	0.00

Table 6: Parameter estimates, standard errors (SE) and 95% confidence intervals (Lower and Upper) from the best-approximating negative binomial regression model relating recipient site shoot counts to the number of days since removal (NumDays) and plot number. Parameter estimates are reported on the log scale.

Parameter	Estimate	SE	Lower	Upper
Intercept	0.205	0.169	-0.126	0.536
NumDays	-0.039	0.251	-0.531	0.454
Plot 3	-0.623	0.259	-1.131	-0.114
Plot 4	-0.588	0.245	-1.068	-0.107
Plot 5	-1.533	0.329	-2.178	-0.888
Plot 6	-1.495	0.357	-2.195	-0.795
$NumDays^2$	-0.213	0.181	-0.568	0.141
$NumDays^3$	0.384	0.167	0.057	0.712
NumDays × Plot 3	-0.364	0.377	-1.103	0.376
NumDays × Plot 4	-0.132	0.375	-0.867	0.603
NumDays × Plot 5	-0.826	0.493	-1.792	0.140
NumDays × Plot 6	-2.671	0.569	-3.787	-1.556
$NumDays^2$ × Plot 3	0.694	0.274	0.157	1.230
$NumDays^2$ × Plot 4	0.637	0.259	0.130	1.145
$NumDays^2$ × Plot 5	1.027	0.335	0.370	1.684
$NumDays^2$ × Plot 6	0.513	0.343	-0.160	1.186
$NumDays^3$ × Plot 3	-0.207	0.247	-0.691	0.277
$NumDays^3$ × Plot 4	-0.216	0.244	-0.694	0.262
$NumDays^3$ × Plot 5	-0.450	0.324	-1.086	0.185
$NumDays^3$ × Plot 6	0.512	0.353	-0.180	1.204
Theta	0.203			

Table 7: Parameter estimates, standard errors (SE) and 95% confidence intervals (Lower and Upper) from the best-approximating logistic regression model relating recipient site percent frequency to the number of days since removal (NumDays) and plot number. Parameter estimates are reported on the logit scale.

Parameter	Estimate	SE	Lower	Upper
Intercept	-0.731	0.042	-0.813	-0.649
NumDays	0.034	0.067	-0.097	0.165
Plot 3	-0.558	0.062	-0.680	-0.435
Plot 4	-0.161	0.060	-0.278	-0.045
Plot 5	-1.665	0.078	-1.819	-1.513
Plot 6	-1.464	0.076	-1.614	-1.316
$NumDays^2$	-0.047	0.046	-0.136	0.043
$NumDays^3$	0.298	0.043	0.213	0.383
NumDays × Plot 3	-0.234	0.099	-0.428	-0.040
NumDays × Plot 4	-0.397	0.096	-0.586	-0.209
NumDays × Plot 5	-1.454	0.140	-1.731	-1.182
NumDays × Plot 6	-1.864	0.143	-2.148	-1.586
$NumDays^2$ × Plot 3	0.179	0.067	0.048	0.309
$NumDays^2$ × Plot 4	0.030	0.065	-0.096	0.157
$NumDays^2$ × Plot 5	0.275	0.074	0.131	0.420
$NumDays^2$ × Plot 6	0.057	0.072	-0.084	0.197
$NumDays^3$ × Plot 3	-0.123	0.062	-0.245	0.000
$NumDays^3$ × Plot 4	0.109	0.062	-0.011	0.230
$NumDays^3$ × Plot 5	0.141	0.083	-0.022	0.305
$NumDays^3$ × Plot 6	0.275	0.085	0.110	0.442